ECE 20875:
Python for Data Science

Summer 2026
Course Syllabus

Instructors

Section 001 (CRN: 38193)
Section WNG (CRN: 14476)
Greg Sturm (gsturm@purdue.edu)
Lectures: MTWRF, 11:00 AM-12:00 PM (Pre-Recorded Online)

See pinned post on Piazza for office hour links.

Prerequisites

CS 15900 (minimum grade of a C-)

Recommended Text

Python Essential Reference, 4th Edition, David M. Beazley, MacMillian, 2009, ISBN No. 978-0672329784

Description: This course will introduce Python programming to students through data science problems. Students will learn Python concepts as well as introductory data science topics, and will use their knowledge of Python (and prior programming experience) to implement data analyses.

Course Outcomes

A student who successfully fulfills the course requirements will have demonstrated:

1. An understanding of regular expressions
2. An ability to use Python to write data analyses
3. An ability to explain when particular data analyses are appropriate
4. An ability to explain the results of data analyses
5. An ability to incorporate classes in their Python programs
6. An ability to incorporate associative arrays in their programs

These outcomes are extremely high level. In more detail, after taking this course you will be able to:

- Write programs in Python that incorporate: basic control structures; functions; data structures such as lists, tuples and associative arrays; classes and objects; higher-order functions; and regular expressions.
- Perform basic scripting tasks (e.g., redirecting the output of one program to the input of another, setting and reading environment variables).
- Understand the basics of sampling, estimation, and hypothesis testing.

- Write data analyses that perform: visualization; textual analysis; regression; classification; and clustering.
- Perform more complex analyses using approaches like neural networks.

Course Assessment: The achievement of the course objectives will be through programming assignments, exams, and a project.

Course Grading
Grades will be assigned as follows:

40% — Exam
25% — 10 programming assignments (all equally weighted)
20% — Project
15% — Quizzes

Grade	Cut-off %
A+	97.0
A	93.0
A-	90.0
B+	87.0
B	83.0
B-	80.0
C+	77.0
C	73.0
C-	70.0
D+	67.0
D	63.0
D-	60.0
F	<60.0

Final grades will be truncated, not rounded up. Final grades may be curved up, but will not be curved down.

Programming Assignments
Programming assignments will be due most weeks. They will test the concepts discussed during the lecture, both programming and statistical. The following rules apply to *all* programming assignments:

1. All programs should run correctly using version 3.13 of the Python interpreter on PyCharm (this should be the default interpreter).
2. Some assignments will include a separate writeup as well as a code submission that will produce the required output when run.
3. Unless otherwise specified, assignments are due at 11:59 PM on the deadline.

Programming assignments will mostly be submitted via GitHub Classroom. The project and a few homework assignments have Gradescope submission parts.

Late submissions will be scaled according to lateness, **docking 20% from your score per day late.**

Oral Exam
If the instructor determines there are substantial similarities between the graded work of two students, or determines the graded work is substantially similar to work created by generative AI, the instructor may replace the graded work in question and/or future graded work with an oral exam(s) for the student(s).

Exams
There will be one closed-book, 1-hour exam administered online, proctored at the regularly scheduled class time. You may bring a one-sided, 8.5”x11” handwritten crib sheet to each exam. Please note “handwritten” does not include notes written on a computer and then printed.

For each exam, you must have:
- Access to a reliable internet connection
- Access to a computer for displaying the exam
- Pen/dark pencil and several pieces of paper for writing answers
- Webcam for exam proctoring

Missing an exam/starting an exam late will result in a grade of 0 for that exam. Exam accommodation will only be granted if you demonstrate that it is absolutely necessary (accompanied by verification) and obtain written approval from me ***in advance*** of the exam. Justification for exam accommodation includes being in a time zone where the exam would begin at a time between 9:00 PM and 7:00 AM local time.

Tentative Course Schedule
Below is the schedule of when and what will be covered in class:

Week	Topics covered	Assignments Due
1 **06/15**	Introduction Git, Python fundamentals Data structures	
2 **06/22**	Higher-order functions Filters and map/reduce List comprehensions, File IO Regular Expressions	**Homework 1 (6/24)**
3 **06/29**	Objects and Classes, Inheritance Histograms Probability distributions and random variables **July 4 (No Lecture)**	**Homework 2 (7/1)** **Homework 3 (7/1)**

Week	Topics covered	Assignments Due
4 **07/06**	Sampling and estimation Hypothesis testing Confidence intervals Matrix Algebra	**Homework 4 (7/8)** **Homework 5 (7/8)**
5 **07/13**	Introduction to supervised learning Linear regression Regularization, Cross validation Exam review **Exam (7/17 at 11:00 AM-12:00 PM EST on Honorlock)**	
6 **07/20**	Natural language processing Introduction to unsupervised learning Clustering: K-Means Clustering: GMMs	**Homework 6 (7/22)** **Homework 7 (7/22)**
7 **07/27**	Classification: KNN Classification: Naive Bayes Classification: Logistic regression Introduction to neural networks	**Homework 8 (7/29)** **Homework 9 (7/29)**
8 **08/03**	Deep learning for visual recognition (CNNs) Conclusion	**Homework 10 (8/5)** **Project (8/5)**